

Assessment questions for PYTHON

Python

1. What is the primary use of the **with** statement in Python?

- a) To define anonymous functions
- b) To handle exceptions
- c) To simplify exception handling during resource management
- d) To create decorators

Answer: c

2. Which of the following statements about Python's GIL is true?

- a) It allows true multi-threading in all Python implementations
- b) It prevents multiple threads from executing Python bytecodes at once
- c) It ensures that Python uses less memory
- d) It is only used in Python 2, not Python 3

Answer: b

3. What is the result of the expression **[] == [] is []**?

- a) True
- b) False
- c) SyntaxError
- d) TypeError

Answer: b

NumPy

1. Which function in NumPy is used to create an identity matrix?

- a) `numpy.eye()`
- b) `numpy.ones()`
- c) `numpy.identity()`
- d) `numpy.diag()`

Answer: a

2. How does NumPy handle broadcasting?

- a) Automatically expands arrays to the same shape
- b) Restricts operations to same shape arrays
- c) Flattens arrays before operations
- d) Throws an error if shapes differ

Answer: a

3. Which of the following returns a view, not a copy?

- a) `arr[:,2]`
- b) `arr.copy()`
- c) `np.array(arr)`
- d) `arr.flatten()`

Answer: a

Pandas

1. Which method combines two DataFrames horizontally by columns?

- a) `merge()`
- b) `concat(axis=1)`
- c) `append()`
- d) `join()`

Answer: b

2. What does `df.loc[0:2]` return?

- a) First 3 rows
- b) First 2 rows
- c) Rows at index 1 and 2
- d) None

Answer: a

3. How do you fill missing values in a DataFrame?

- a) `df.fill()`
- b) `df.replace()`
- c) `df.fillna()`
- d) `df.impute()`

Answer: c

Machine Learning

1. What does regularization do in ML?

- a) Reduces training time
- b) Adds penalty to prevent overfitting
- c) Normalizes input data
- d) Converts non-linear to linear models

Answer: b

2. **Which is used to evaluate classification models?**

- a) MAE
- b) R-squared
- c) F1-score
- d) MSE

Answer: c

3. **Best strategy for handling imbalanced datasets?**

- a) Normalization
- b) Drop missing values
- c) Resampling
- d) PCA

Answer: c

Artificial Intelligence

1. **What does the minimax algorithm do?**

- a) Maximizes the heuristic score
- b) Minimizes computation
- c) Finds optimal move in adversarial games
- d) Solves linear equations

Answer: c

2. **Real-world example of expert systems?**

- a) Google Translate
- b) Medical diagnosis
- c) Game playing
- d) Image generation

Answer: b

3. **Role of heuristic in AI?**

- a) Guarantees optimal solution
- b) Estimates cost to reach goal
- c) Maximizes accuracy
- d) Creates neural networks

Answer: b

SciPy

1. **Which module is used for optimization?**

- a) `scipy.linalg`

- b) `scipy.signal`
- c) `scipy.optimize`
- d) `scipy.stats`

Answer: c

2. **Solve system of equations $Ax = b$ using SciPy:**

- a) `solve(A, b)`
- b) `np.linalg.solve(A, b)`
- c) `scipy.solve(A, b)`
- d) `optimize.solve(A, b)`

Answer: a

3. **Purpose of `scipy.interpolate`?**

- a) Matrix factorization
- b) Statistical tests
- c) Signal denoising
- d) Estimating intermediate values

Answer: d

Keras (Deep Learning)

1. **Which callback prevents overfitting in Keras?**

- a) `ModelCheckpoint`
- b) `EarlyStopping`
- c) `LambdaCallback`
- d) `TensorBoard`

Answer: b

2. **Output of `model.fit()` in Keras?**

- a) Model weights
- b) Training loss
- c) History object
- d) Prediction

Answer: c

3. **What does dropout do?**

- a) Normalizes inputs
- b) Adds regularization
- c) Randomly disables neurons during training
- d) Increases learning rate

Answer: c

LLMs (Large Language Models)

1. **Purpose of positional encoding in transformers?**

- a) Reduce memory
- b) Track word importance
- c) Encode token order
- d) Tokenize input text

Answer: c

2. **Which mechanism lets transformers focus on relevant parts?**

- a) Pooling
- b) Dropout
- c) Self-attention
- d) Backpropagation

Answer: c

3. **What is fine-tuning in LLMs?**

- a) Pretraining from scratch
- b) Tokenizing data
- c) Adapting pretrained model to task
- d) Changing architecture

Answer: c